AP CALCULUS AB

Unit 1 • Worksheet 1-8A

Worksheet 1-8A • MVT

Name: _____ Date: _____

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

For $f(x) = x^2$ on $[0, 4]$, which value of c satisfies the MVT?

(A) 1
(B) 2
(C) 3
(D) 4

Question 2

NO CALCULATOR

For which function does the MVT apply on $[-1, 1]$?

(A) $|x|$
(B) $\frac{1}{x}$
(C) x^3
(D) $\sqrt[3]{x}$

Question 3

NO CALCULATOR

f is differentiable, $f(1) = 4$ and $f(5) = -4$. Which must be true?

(A) $f'(c) = -2$ for some c in $(1, 5)$
(B) $f'(3) = -2$
(C) $f(c) = 0$ only once
(D) $f'(c) = 0$ for some c

Question 4 (Free response)

NO CALCULATOR

Find all values of c guaranteed by the MVT. (Use $g'(x) = 3x^2$ and $k'(x) = -\frac{1}{x^2}$.) (a) $g(x) = x^3$ on $[0, 3]$ (b) $k(x) = \frac{1}{x}$ on $[1, 4]$ (c) $g(x) = x^3$ on $[-2, 2]$

Question 5 (Free response) NO CALCULATOR

A car's position $s(t)$ (miles) is differentiable; $s(0) = 0$, $s(1.5) = 90$, $s(3) = 165$.

(a) Must the car's velocity have been exactly 60 mph at some time in $(0, 1.5)$? Justify.
(b) Must it have been exactly 50 mph at some time in $(1.5, 3)$? Justify.